

Date: 22-1-25

Lec # 06

Day: M T W T F S

Partial derivatives:-

Rule for finding partial derivative.
tells the slope of tangent line.

$$* \quad z = 4x^2 + 3y^3$$

Solution:-

$$\frac{\partial z}{\partial y}$$

$$z_x = \frac{\partial z}{\partial x} = 8x + 0$$

$$z_y = \frac{\partial z}{\partial y} = 0 + 9y^2$$

Exp ①:- $z = 4x^2y^2 + 12x^3y$

Solution:-

$$z_x = \frac{\partial z}{\partial x} = 8xy^2 + 36x^2y$$

$$z_y = \frac{\partial z}{\partial y} = 8x^2y + 12x^3$$

Ans -

- ① To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x .
- ② To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y .

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Exp:- If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find f_x , f_y , $f_x(2, 1)$, $f_y(2, 1)$.

Solution:-

$$1) f_x = \frac{\partial f}{\partial x} = 3x^2 + 2xy^3 - 0$$

$$2) f_y = \frac{\partial f}{\partial y} = 0 + 3x^2y^2 - 4y$$

$$\begin{aligned} 3) f_x(2, 1) &= 3(2)^2 + 2(2)(1)^3 \\ &= 3(4) + 4(1) \\ &= 12 + 4 = \boxed{16} \end{aligned}$$

$$\begin{aligned} 4) f_y(2, 1) &= 3(2)^2(1)^2 - 4(1) \\ &= 3(4)(1) - 4 \\ &= 12 - 4 = \boxed{8} \quad \text{Ans.} \end{aligned}$$

Exp:- If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$

Solution:-

$$f(x, y) = \sin\left(\frac{x}{1+y}\right)$$

$$\frac{\partial f}{\partial x} = f_x = \cos\left(\frac{x}{1+y}\right) \times \frac{\partial}{\partial x} \left(\frac{x}{1+y}\right)$$

$$f_x = \cos\left(\frac{x}{1+y}\right) \times \frac{1}{1+y}$$

$$\frac{\partial f}{\partial y} = f_y = \cos\left(\frac{x}{1+y}\right) \times \frac{\partial}{\partial y} \left(\frac{x}{1+y}\right)$$

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formula:- $\left(\frac{u}{v} \right)' = \frac{u'v - v'u}{v^2}$

$$f_y = \cos \left(\frac{x}{1+y} \right) \times \left[\frac{(0)(1+y) - (1)(x)}{(1+y)^2} \right]$$

$$f_y = \cos \left(\frac{x}{1+y} \right) \left[\frac{-x}{(1+y)^2} \right] \text{ Ans.}$$

Exp:- $f(x, y) = 4 - x^2 - 2y^2$ find f_x , f_y , $f_x(1, 1)$ and $f_y(1, 1)$.

Solution:-

$$1) \quad f_x = \frac{\partial f}{\partial x} = 0 - 2x - 0$$

$$2) \quad f_y = \frac{\partial f}{\partial y} = 0 - 0 - 4y$$

$$3) \quad f_x(1, 1) = -2(1) \\ = \boxed{-2}$$

$$4) \quad f_y(1, 1) = -4(1) \\ = \boxed{-4} \quad \text{Ans.}$$